

Question number	Answer	Additional guidance	Mark
(a)	A calculation showing the following steps: - determination of tidal volume (1) - calculation of volume of air in 1 min (1) - calculation of volume of air in one hour (1)	$(500-150) = 350$ $350 \times 12 = 4.2 \text{ (dm}^3 \text{ min}^{-1}\text{)}$ Accept 4200 (cm³ min⁻¹) $(4.2 \times 60) = 250 \text{ dm}^3 \text{ hour}^{-1}$ Accept $(4200 \times 60) = 250000$ cm³ (hour⁻¹) max 2 marks if no units / or incorrect units accept answer in standard form 2.5×10^5	(3)

Question number	Answer	Additional guidance	Mark
(b)(i)	An answer that includes the following point: (cardiac) diastole (1)		(1)

Question number	Answer	Additional guidance	Mark
(b)(ii)	100	accept 1.67 beats per second accept range 100 - 125	(1)

Question number	Answer	Additional guidance	Mark
(b)(iii)	An answer that includes three of the following points: Similarity - both traces have the same {PQRST/ described} wave (1) - R wave of similar height (1) Differences - normal ecg shows a lower heart rate (1) - in the patient trace the gap between T & P is shorter (1)	both show clear PQRST wave accept converse accept converse	(3)

Question number	Answer	Additional guidance	Mark
)	An answer that includes four of the following points: {identification / isolation} of (corrected) AAT gene (1) - description of one enzyme used to insert gene (1) - adenoviruses can enter host cell (1) - and combine their own genetic material with that of the cell (1) - therefore a functional AAT protein can be produced (1)	E.g. cut the adenovirus DNA using the same restriction endonuclease / insert the gene into the adenovirus DNA using DNA ligase adenovirus acts as a vector	(4)